aUToronto

R2Y5 0-0-0 Challenge
Sensor Suite Optimization

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For the R2Y5 competition, all teams had to engineer a sensor suite for the car.

There's a problem...

These sensor suites end up being impractical!

Optimize the sensor suite of your car.

R2Y5 RULESET

Extra rules we have to consider in addition to our problem statement

Compliance with specific regulations

FMVSS111, FMVSS127, FMVSS208

Compliance with regulations in 5 states/provinces

California, Michigan, Texas, Arizona, Ontario

Benchmark analysis with another robotaxi

We have chosen Waymo and Tesla

Assumption: We will be optimizing for regulatory requirements and sensor cost optimization

AGENDA

Problem statement → Background information → MVP → Regulations → Self-certifications → Key learnings

Background Information

Problem statement

Background information

MVP

Regulations

Self-certifications

Key learnings

Sensor Modalities

Cameras

2D image capturing for recognizing road signs, lane markings, and colors

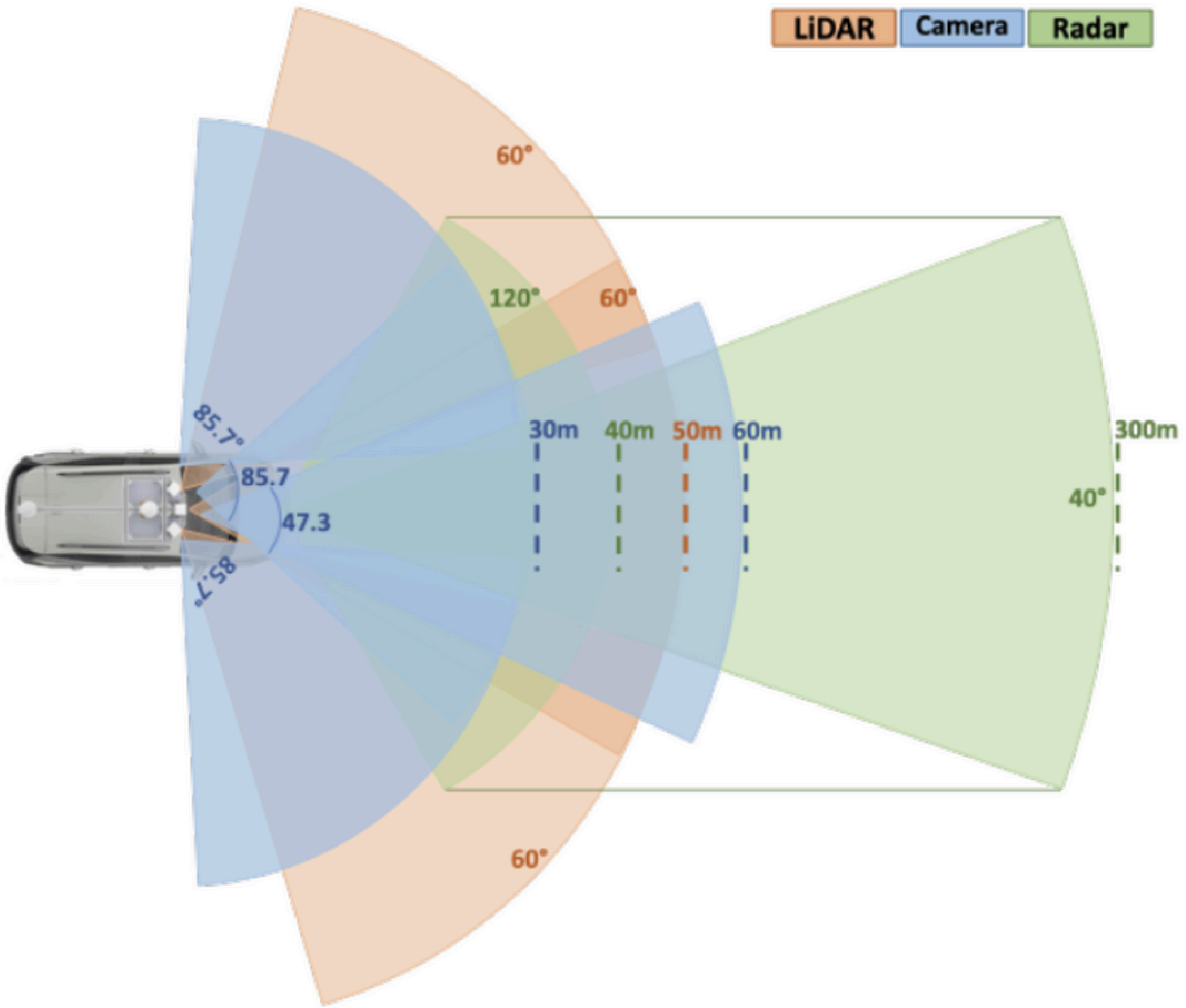
LiDAR

Detects obstacles, road features, and pedestrians using 3D point clouds

Radar

Tracks objects and their speed using doppler effects

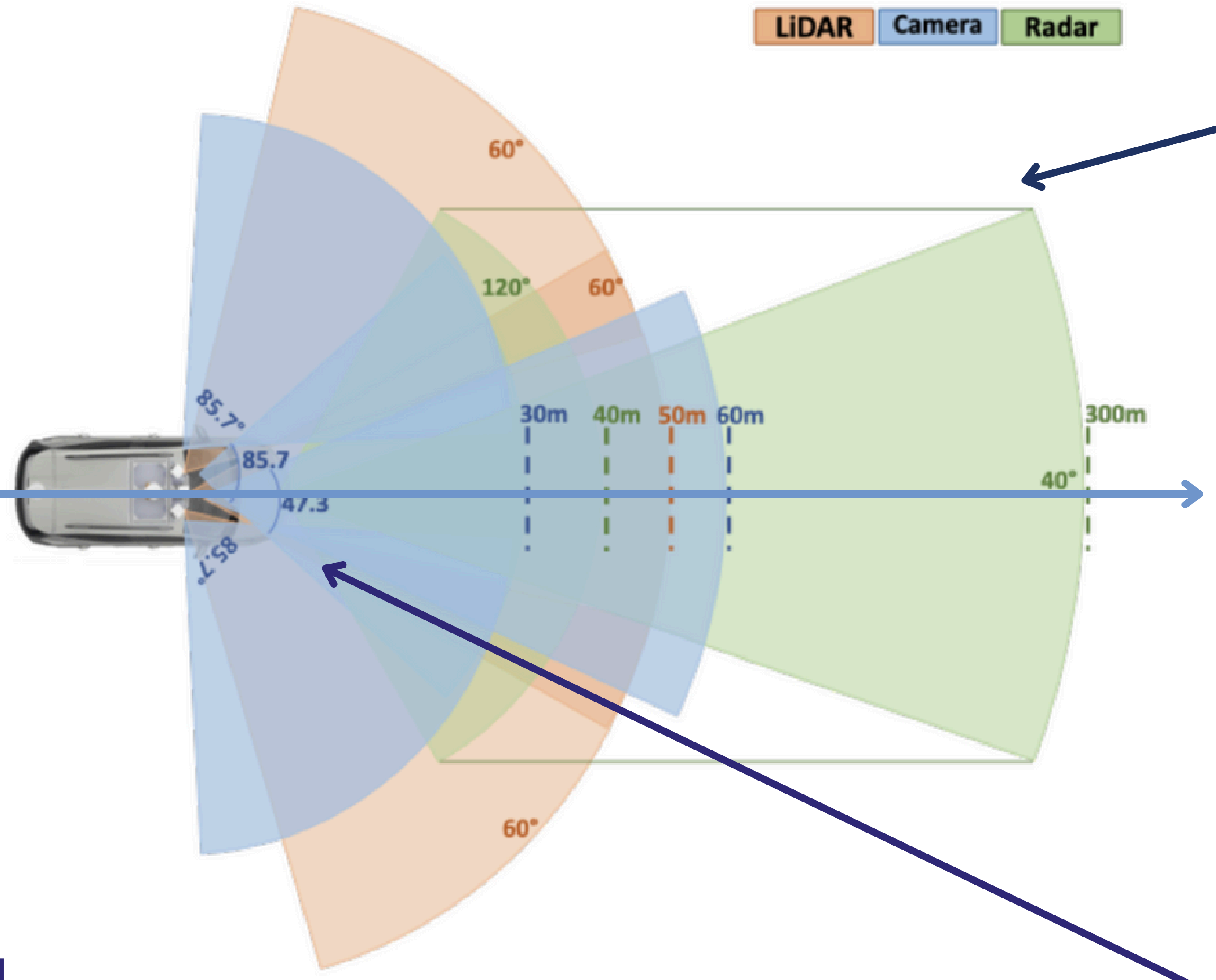
Our R2Y5 Sensor Suite



[1]

A lot of room for improvement...

Completely no coverage on the back of the car



[1]

300 meter range is not needed

Lots of overlap

OUR SCENARIO & ASSUMPTION

We comply with the standards

Context

Our focus for regulatory compliance will be on the **sensors**, and not the autonomous car as a whole.

Physical Car

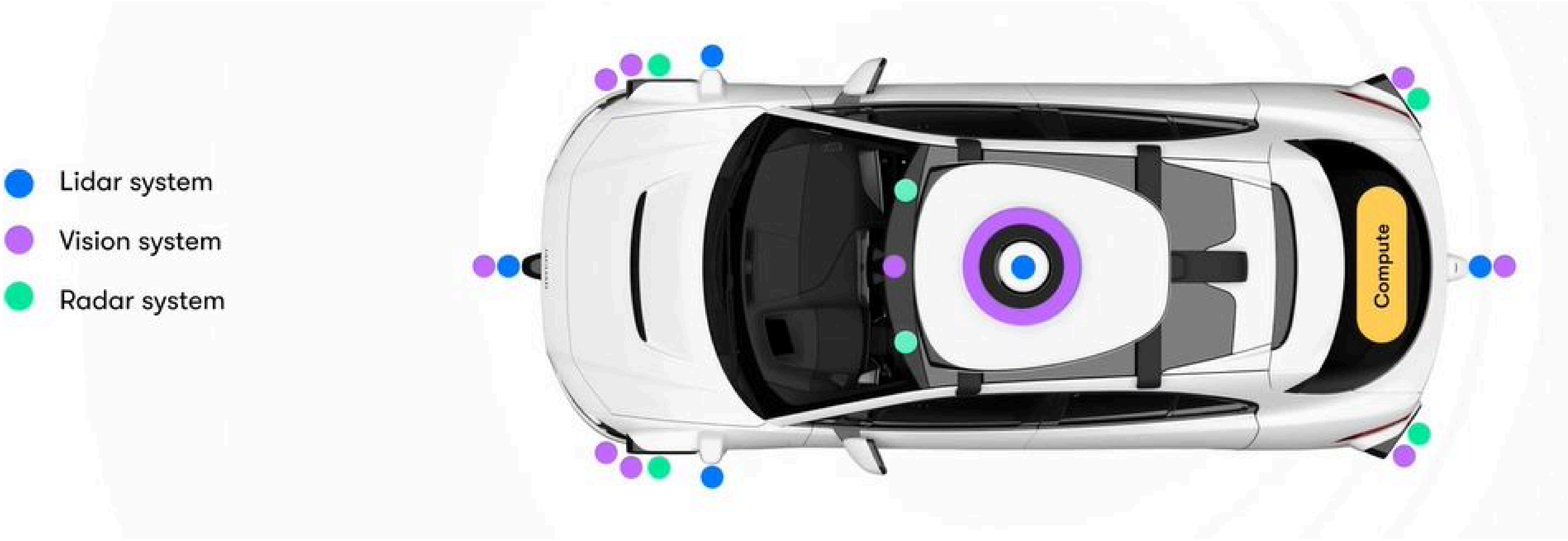
We will be assuming that the underlying car (Chevrolet EUV Bolt) already complies with all the regulations.

MVP

Problem statement
Background information
MVP
Regulations
Self-Certifications
Key learnings

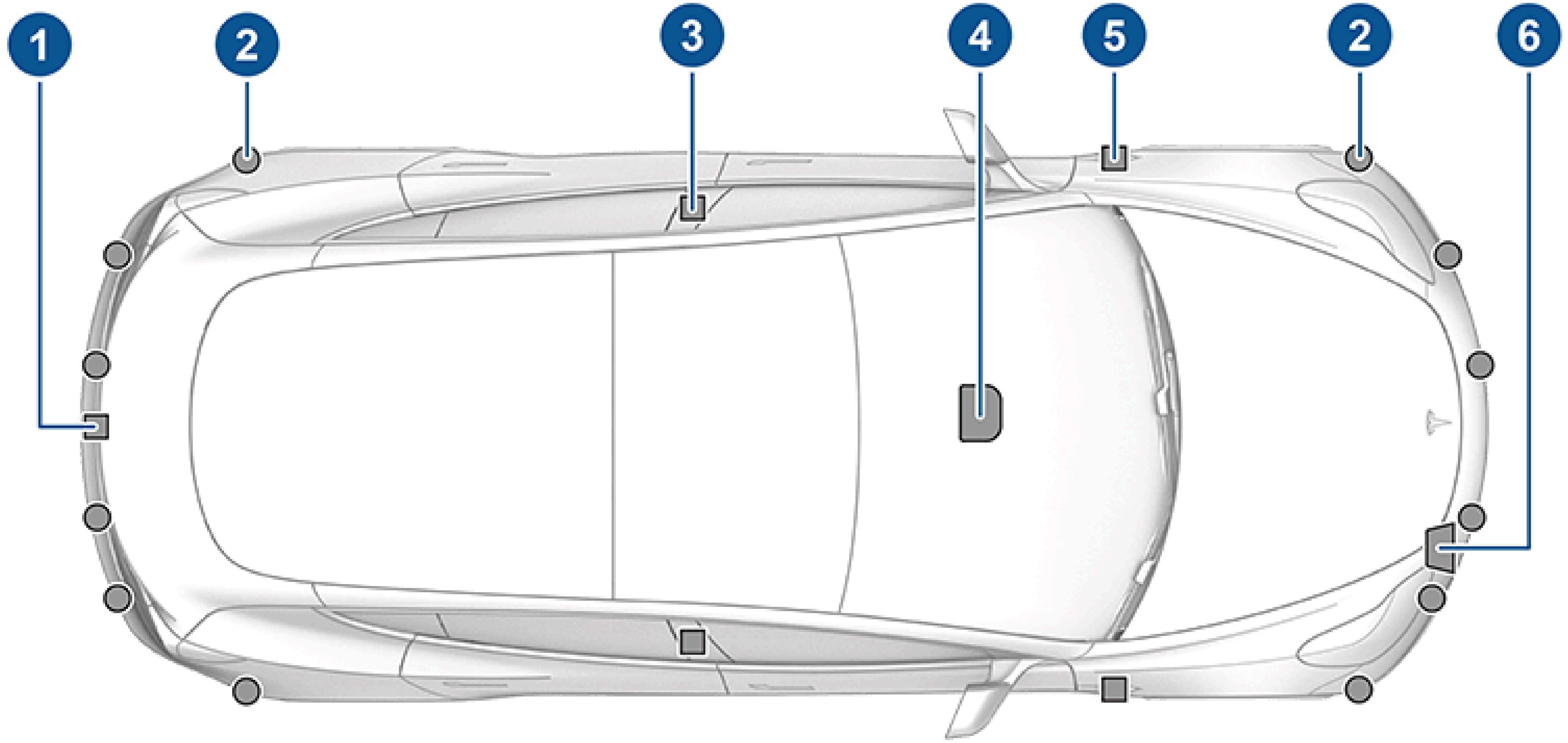
INSPIRATION

Waymo's sensor suite: [1]



INSPIRATION

Tesla’s sensor suite:
#6 is radar, #2 is ultrasonic [2]



DESIGN PRINCIPLES AND CONSTRAINTS

Camera heavy for cost and power efficiency.

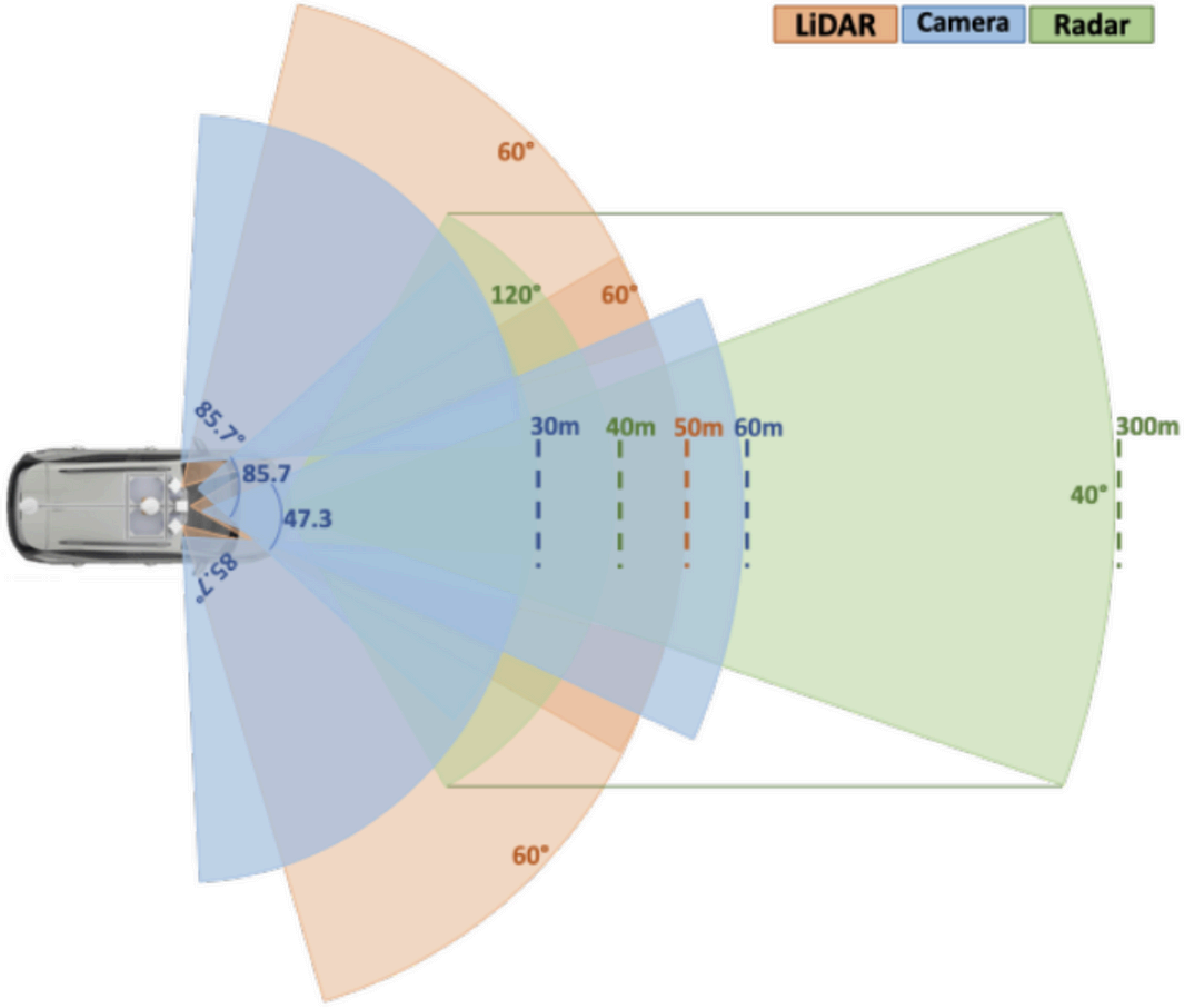
Having minimal sensor overlap, but not decreasing functionality.

4-5

Leveraging LiDAR and radar to target L4-L5 autonomy.

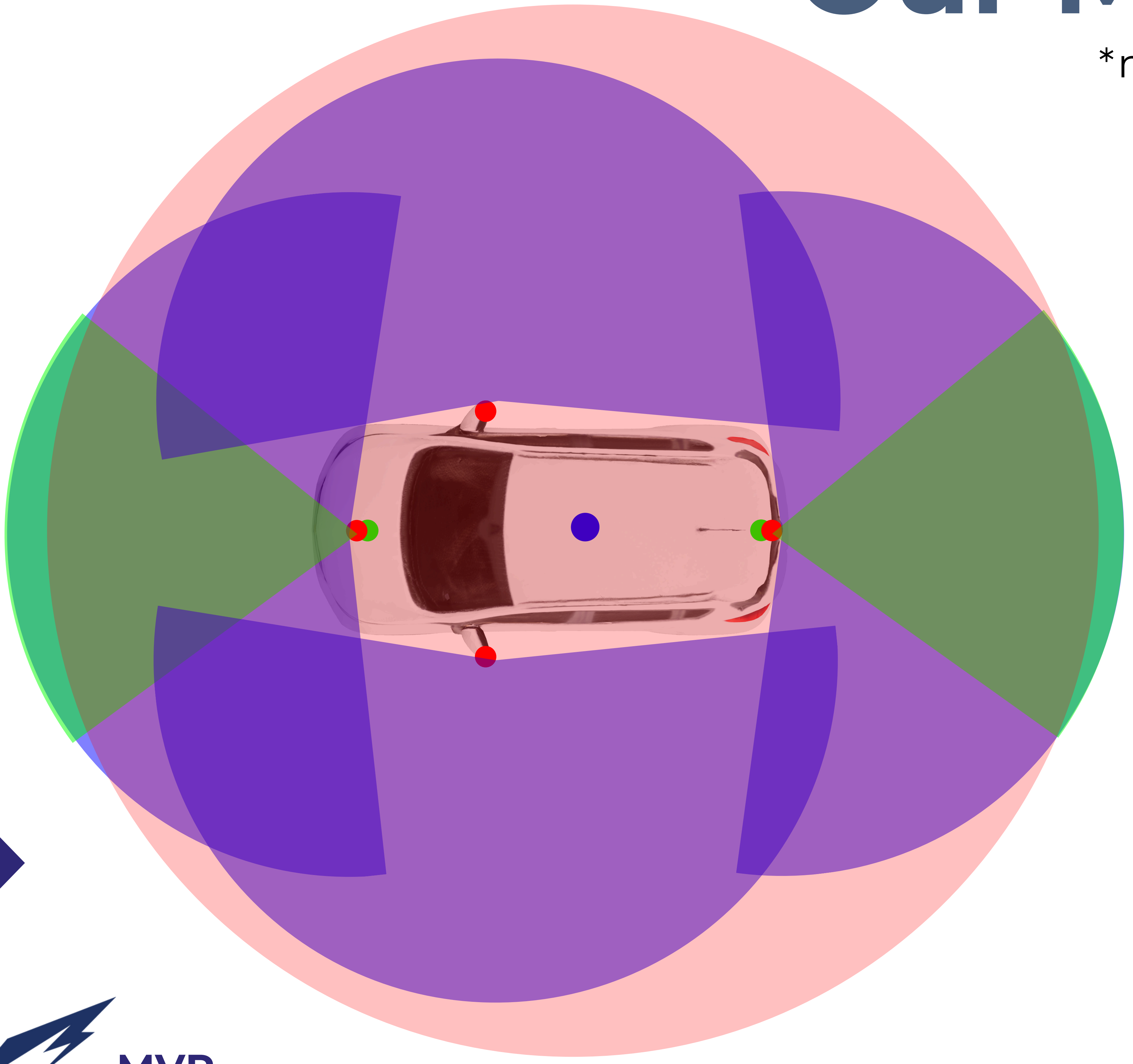
Compliant with all the regulations in our chosen locations.

Before


LiDAR
Camera
Radar
60°
120°
60°
85.7°
85.7
47.3
85.7°
30m
40m
50m
60m
300m
40°
60°

Our MVP

*not to exact scale



Lidar
200 m range

Radar
90 m range

Camera

Our MVP

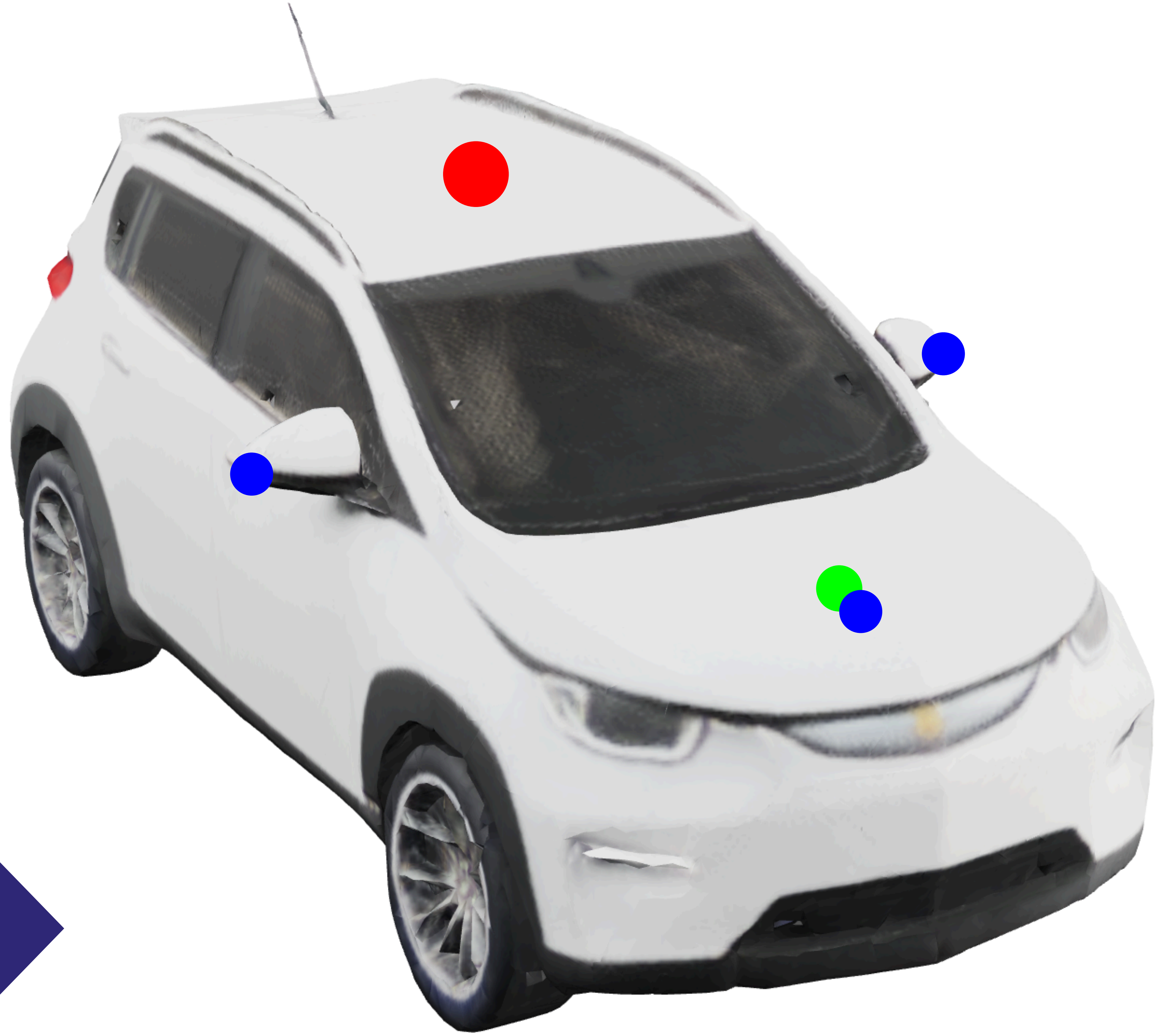
Our MVP

Lidar

Radar

Camera

Our MVP

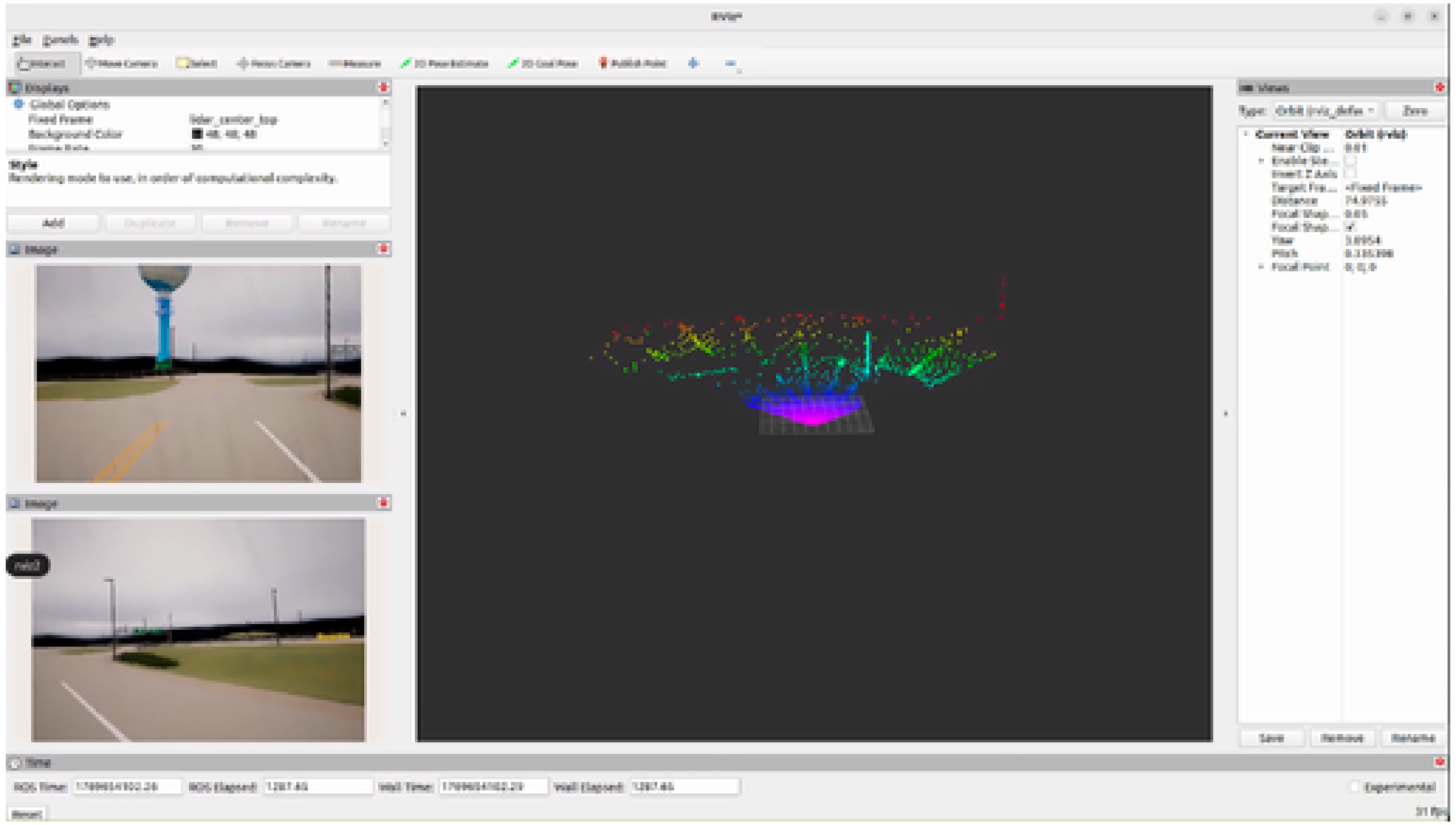


Extra Additional Features:

1. All the sensor data 1 minute before a crash is saved.

2. There will be an internal alerting system to indicate to the driver to take control again.

Simulation Testing



Regulatory Compliance

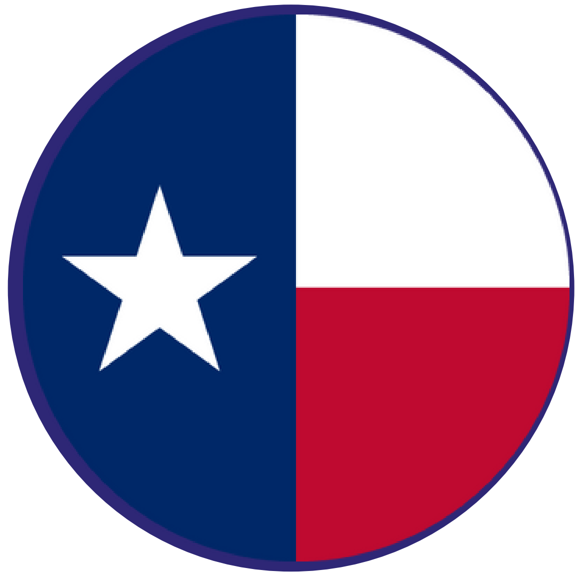
Problem statement → Background information → MVP → Regulations → Self-Certifications → Key learnings

OUR STATES & PROVINCE

Required States

California, USA

Michigan, USA



Texas, USA

Additional 2

Arizona, USA

Toronto, CAN

MAIN REGULATORY BODIES

[3]

[2]

FEDERAL MOTOR VEHICLE SAFETY STANDARDS (FMVSS)

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset] [4]

[3]

FMVSS 111: Regulation for **Rear Visibility Requirements,** including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

FMVSS 127: Regulation for **Automatic Emergency Braking** and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

FMVSS 208: Regulation for **Occupant Crash Protection** design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.

DMV REGULATIONS

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset] [4]

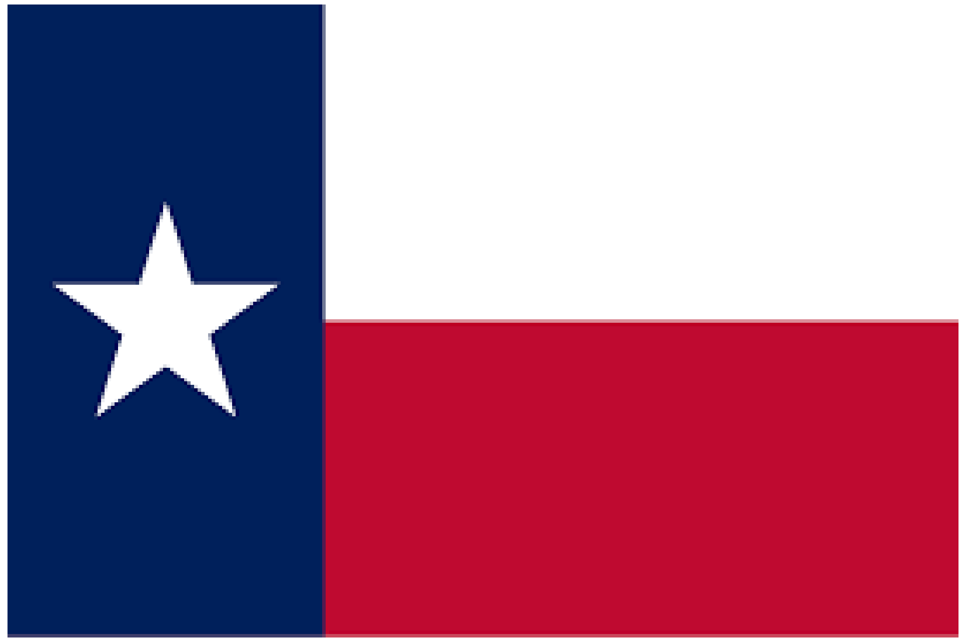
1. Provide an easily accessible method for the driver to engage or disengage the autonomous driving system.
2. Include an in-vehicle indicator that informs the driver when the autonomous driving system is active.
3. Have a system that alerts the driver if the autonomous driving system fails and either allows manual takeover or safely brings the vehicle to a complete stop.
4. Comply with all applicable Federal Motor Vehicle Safety Standards.
5. Include a separate mechanism to record and store sensor data from at least 30 seconds before a collision.

STATE OF CALIFORNIA & TEXAS

Both California and Texas use DMV permits to grants manufacturer access to operate their AVs on public roads.

Clearing all these requirements is enough for California

- There are additional requirements for Texas:
  - Need to create a first responders safety plan (more on our self certification soon!)
  - The DMV must have enhanced authority to oversee autonomous vehicle operations
  - Need to report safety data and incidents [20]



STATE OF ARIZONA & MICHIGAN

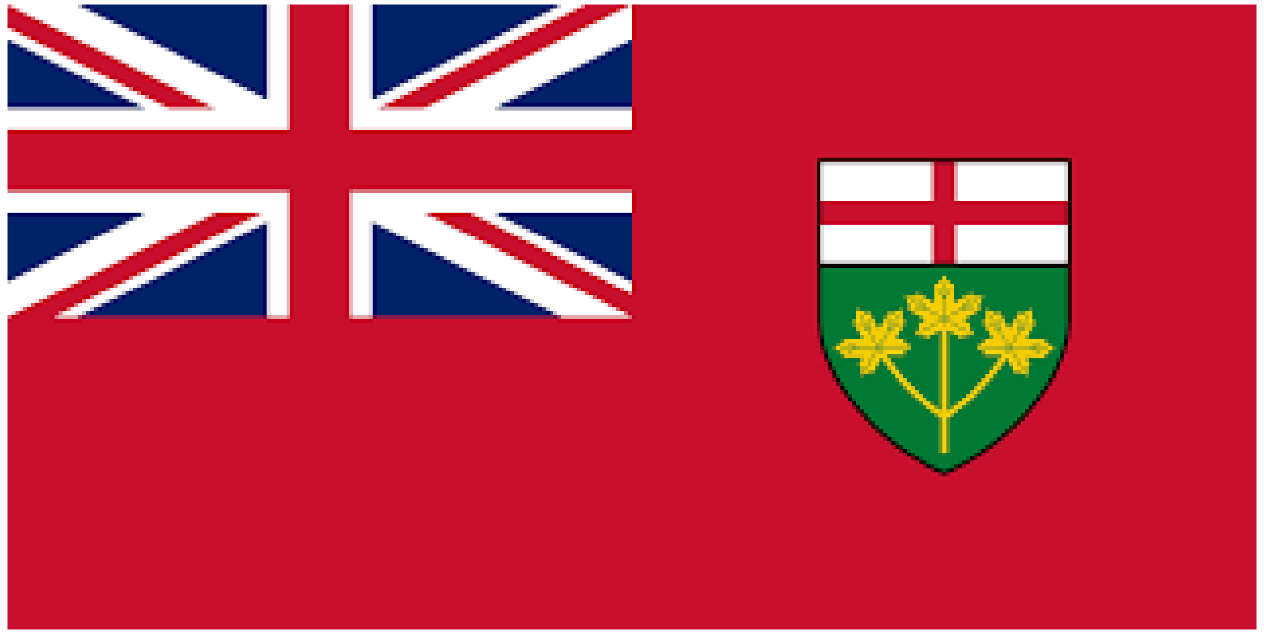
Does not require a DMV permit, but require other specific compliance requirements

1. Must submit a plan to ADOT and the Arizona Department of Public Safety
   - explains how law enforcement should interact with your autonomous vehicles during emergencies, traffic stops, and accidents.
2. A self certification acknowledging:
   - Compliance with all federal safety standards.
   - Compliance with all Arizona traffic and motor vehicle safety laws. [23, 24, 25]
   - can achieve a “minimal risk condition” (i.e., safely stopping) in case of system failure.

1. Must obtain special manufacturer plates issued by the Michigan Secretary of State.
2. Must report proof of insurance, business presence in Michigan, and details about the autonomous system to the Secretary of State. [21, 22]

PROVINCE OF ONTARIO

In Ontario, fully operational AVs are not allowed. We are only allowed to test. In order to test, we must join the Automated Vehicle (AV) Pilot Project.

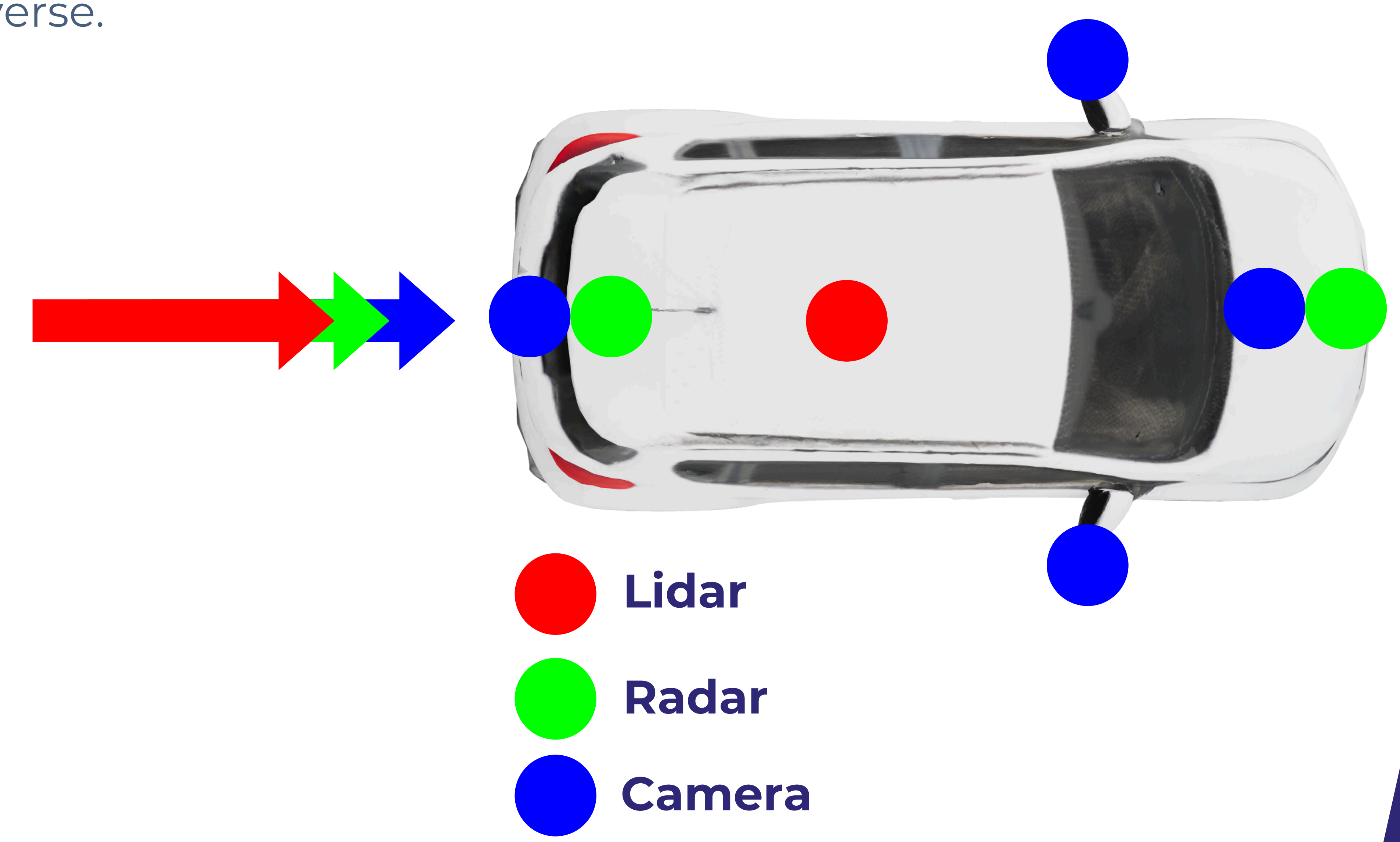


1. Safety & Cybersecurity: Operators must submit a plan outlining safety and cybersecurity measures taken for risk mitigation
2. All participating vehicles must comply with the Motor Vehicle Safety Act. [26, 27, 28]
    a. File annual reports on testing data
    b. Cooperate in six-monthly program reviews with MTO

FMVSS 111: REAR VISIBILITY

FMVSS 111: Regulation for Rear Visibility Requirements, including equipment of rearview **camera** that provides a clear field of view behind the vehicle when in reverse.

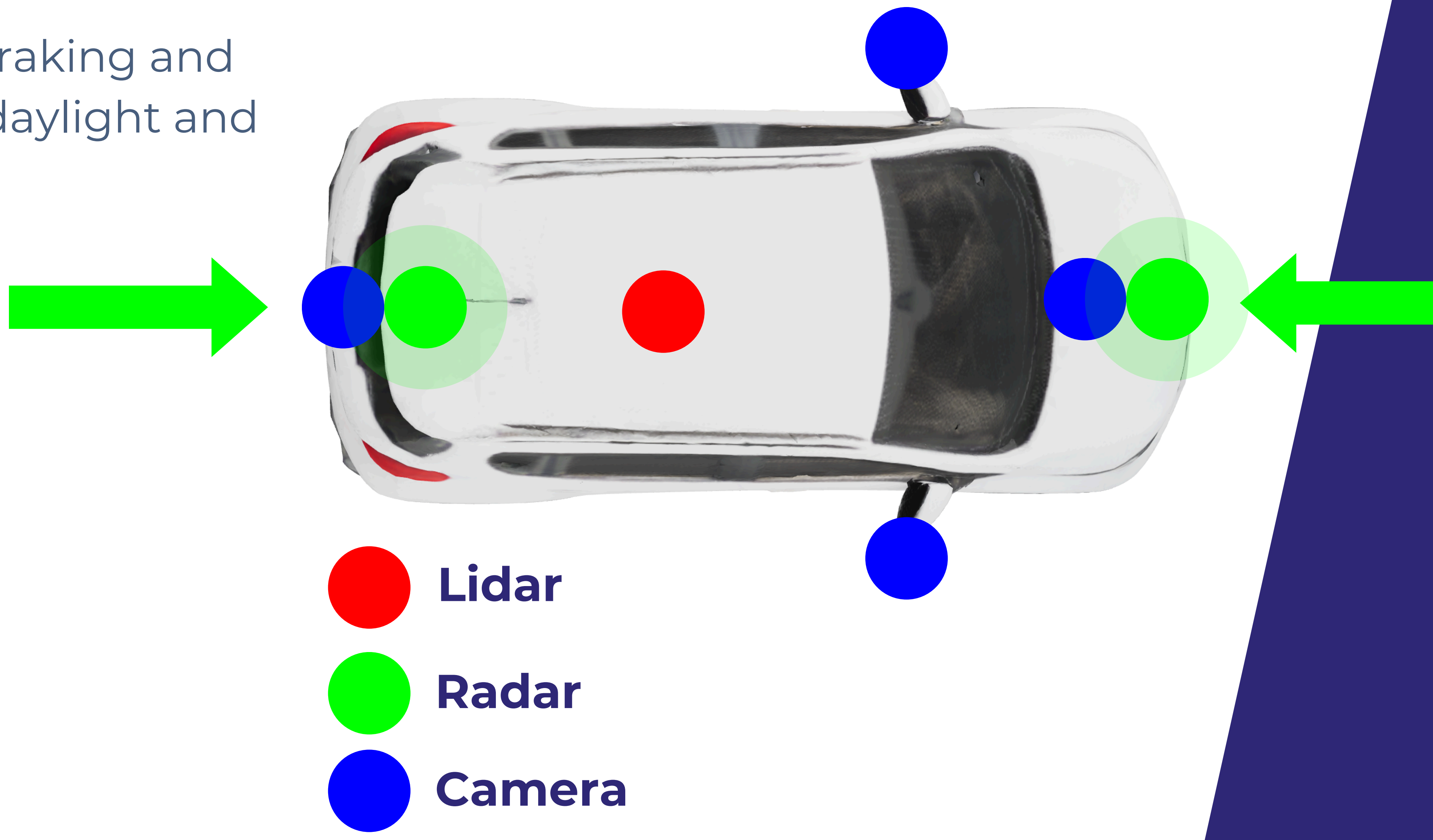
Our sensor suite has 3 modalities of sensors for rear visibility which is more than enough to comply with this regulation, as the minimum requirement is just a clear field of view.



FMVSS 127: AUTOMATIC EMERGENCY BRAKING

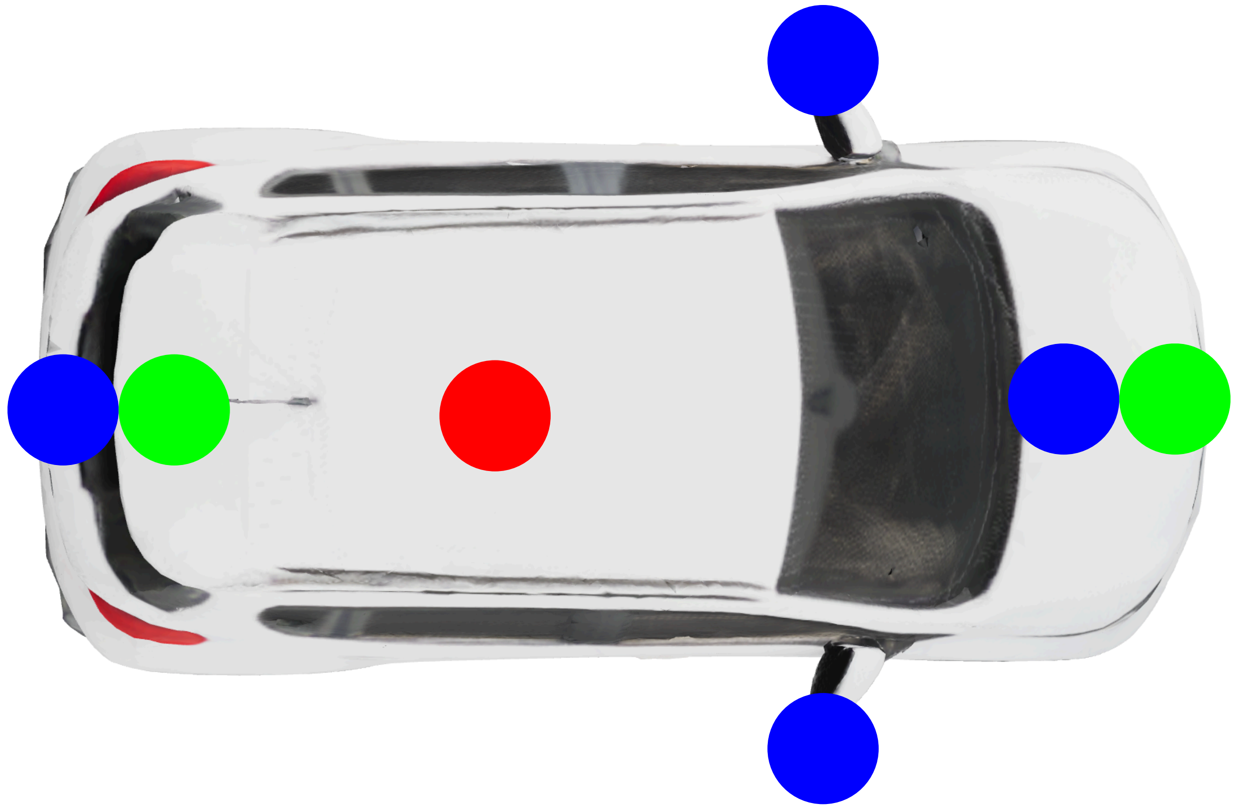
FMVSS 127: Regulation for Automatic Emergency Braking and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

Radar sensors are best for detection during darker/misty/low visibility scenarios, and we have 75 degrees facing the front and back.



FMVSS 208: OCCUPANT CRASH PROTECTION

FMVSS 208: Regulation for Occupant Crash Protection design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.



Our sensor suite has nothing to do with this regulation, as its regarding the functionality of the car itself. By our assumption, we will comply with this regulation.

Our Sensors

Valeo cameras 4x [7]

Brand	H x V FOV (degrees)	Price	Resolution (MP)	Power (W)
Valeo	195 x 155	21	1	<2.5

Hella radars 2x [7]

Brand	Model	Range (m)	H x V FOV (degrees)	Price	Power (W)
Hella	SRR5G3	90	75 x 10	35	22.5

Ouster OS1 LiDAR [12], [13]

Brand	Model	Range (m)	H x V RES (degrees)	H x V FOV (degrees)	Price	Power (W)
Ouster	OS1	200	0.18 x 0.7	360 x 42.4	~2000	14-20

OS1 LiDAR cost

$2400 (listed price on a website)
x
85% (bulk discount assumption) =

~$2000 (bulk purchase price)

KEY SPECS

Total Cost of Sensors:
$2154

Total Energy Consumption: **72W**

Old Mileage:
259 miles

New Mileage:
231 Miles

Comparing our MVP

[8], [9], [10], [11]

Vehicle	Sensor Suite Power (W)	Suite kWh/mile (est.)	Vehicle kWh/mile (base)	Combined kWh/mile (approx.)
Tesla Model 3	230-300	0.005	0.24	0.245
Waymo (Pacific PHEV)	2,500-3,000	0.08	0.33	0.41
Bolt EUV + 72 W Sensor (30 mph)	72	0.0024	0.28	0.2824
Bolt EUV + 72 W Sensor (60 mph)	72	0.0012	0.28	0.2812

Self-Certification Plan & ODD Sensor Compliance

Problem statement → Background information → Regulations → MVP → Self-Certifications → Key learnings

ODD HIGH LEVEL OVERVIEW [14, p.42]

Environmental

Weather
Geolocation
Object Avoidance

Public Roads & Traffic information

Parking
Road Types
Congestion
Road Grade

Autonomy

Path Planning & Execution x2
Lateral Controls
Longitudinal Controls x2

CONNECTION TO MVP SENSOR SUITE

ODD Requirements

what must the ODD do?

what must the sensor do?

↓

Sensor Functional Requirements

↓

where?

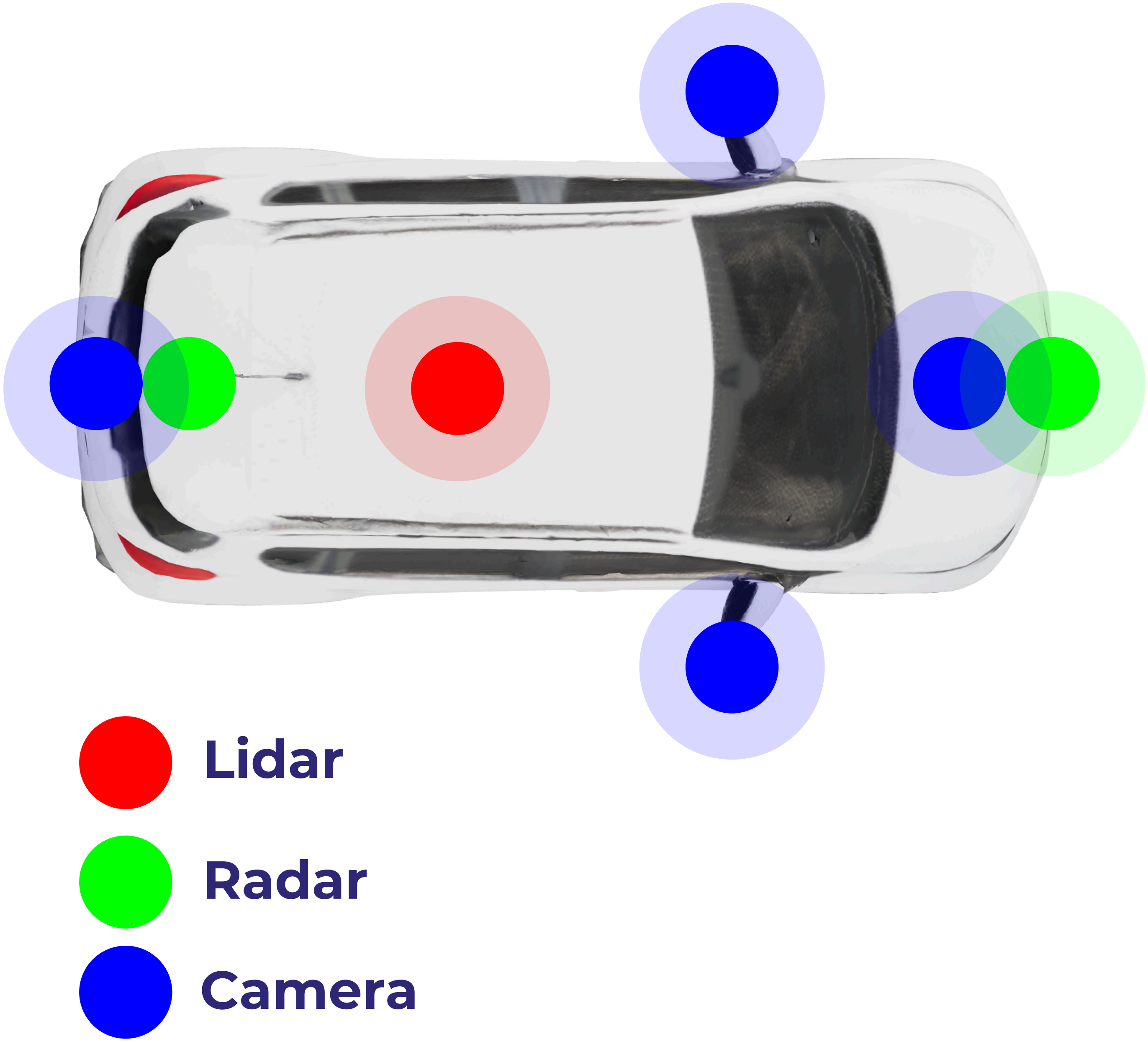
Sensor Coverage

↓

metrics used for pass/fail criteria

Test and Certification Evidence

ENVIRONMENTAL [16, pp. 131-165] [17] [18]



Summarized ODD Requirements

Vehicle recognition, localization, and avoidance of obstacles under claimed environmental conditions.

Sensor Functional Requirements

Provide front coverage for roadway objects within 40 m and ±60° using camera/LiDAR/RADAR

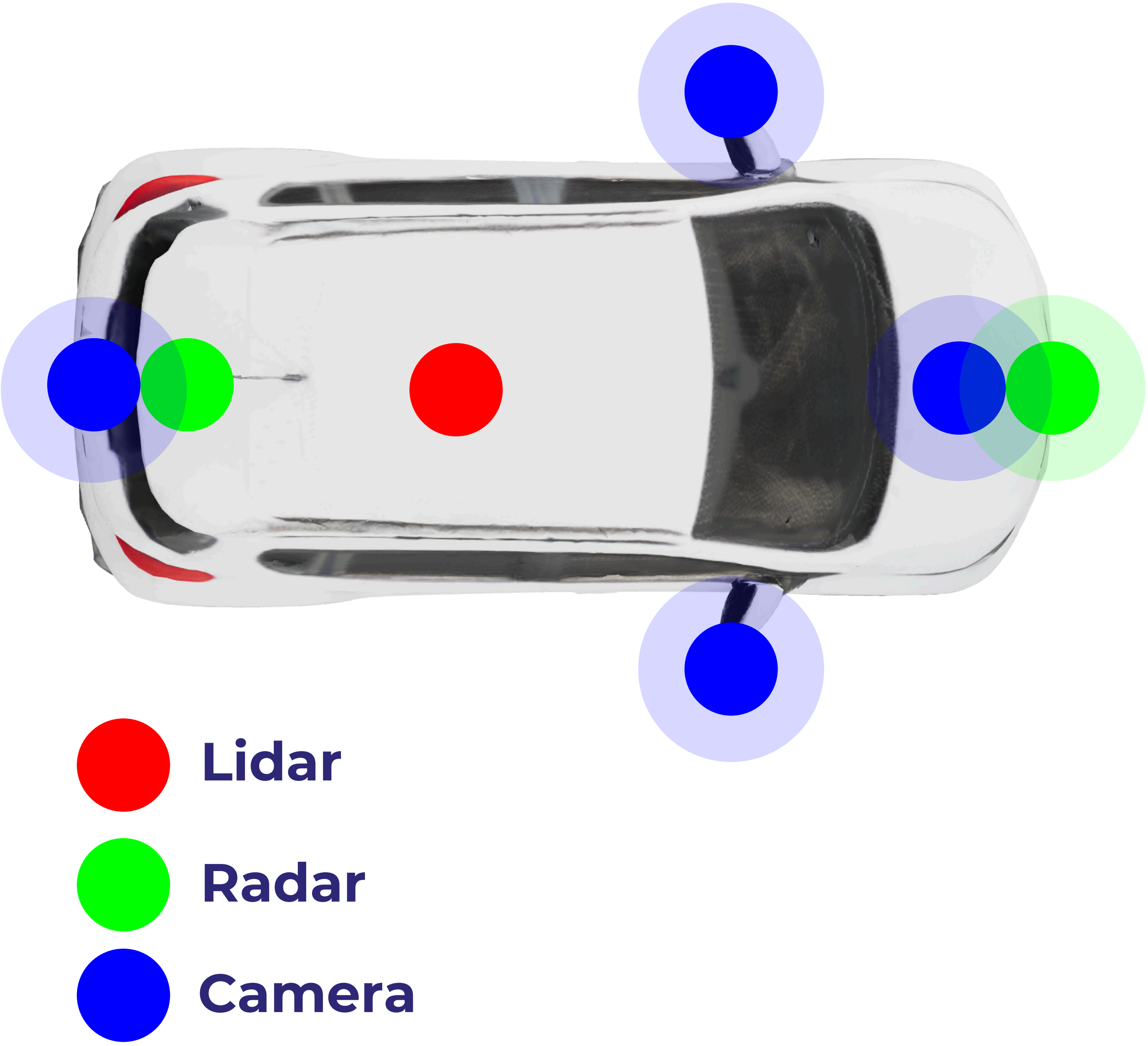
Detect lane lines across any time of day + any weather

Support localization accuracy targets under GPS attenuation >63 dB and detect recovery in ≤1 s

Test & Certification through Simulation, Closed-Track Testing, Real-World Testing

PUBLIC ROADS, TRAFFIC INFORMATION

[16, pp. 131-165] [17] [18]



Summarized ODD Requirements

Demonstrate law-abiding and safe operation on public roads across all claimed roadway conditions.

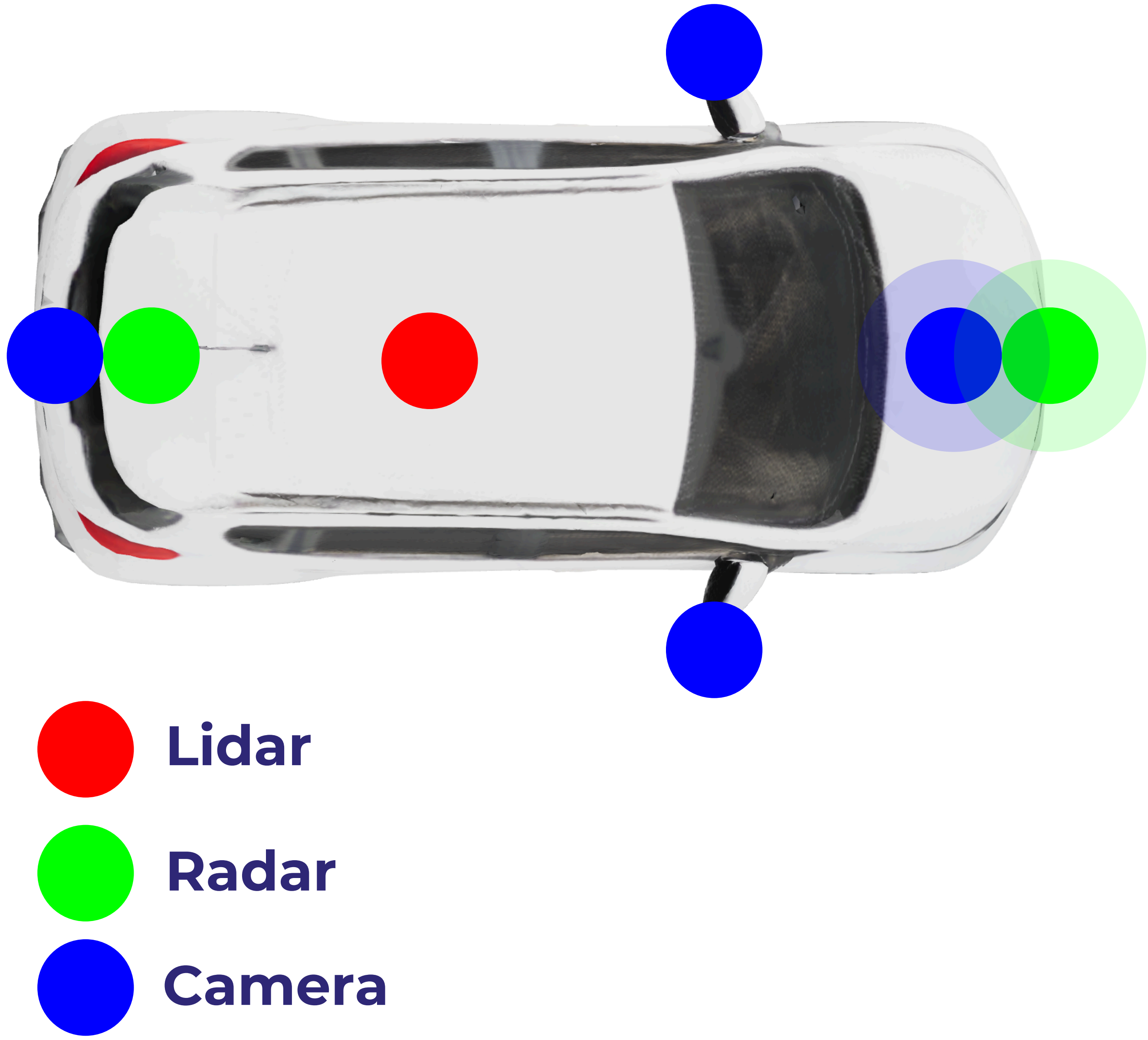
Sensor Functional Requirements

Lane line detection distance, occlusion tolerance, and road type capability

Congestion scenarios rely on tracking surrounding vehicles

Test & Certification through Simulation, Closed-Track Testing, Real-World Testing

AUTONOMY [16, pp. 131-165] [17] [18]



Summarized ODD Requirements

Demonstrate the vehicle can plan, decide, and execute driving actions autonomously and safely.

Sensor Functional Requirements

Track traffic lights and meet strict latency requirements

Follow traffic light state behaviors

Emergency vehicle response must be triggered by emergency light detection and executed safely

Explicitly logs detection and maneuver execution windows

Test & Certification through Simulation, Closed-Track Testing, Real-World Testing

How did we connect ODD-Level Tests to System Certification?

SELF-CERTIFICATION PLAN [14]

We validate system compliance with:

FMVSS 111 FMVSS 127 FMVSS 208

We validate operational compliance with the following ODDs:

Environmental

Weather
Geolocation
Object Avoidance

Public roads & Traffic information

Parking
Road Types
Congestion
Road Grade

Autonomy

Path Planning & Execution
Lateral Controls
Longitudinal Controls

This certification is supported by:

- Verified ODD-level test evidence
- Sensor coverage and performance validation
- Simulation, track, and real-world testing

CHALLENGES & KEY LEARNINGS

Translating Regulations to Sensors.

FMVSS regs are written for functions, not sensors. None of them say “use LiDAR” or “use radar.”

Benchmarking against real Robotaxis.

Waymo/Cruise/Zoox aren’t just overbuilt, they’re validation-driven.

Self-certification without third-party validation.

Self-certification shifts responsibility to the developer.

THANK YOU

QUESTIONS?